NCEA Level 1 Mathematics and Statistics

91947 · Demonstrate mathematical reasoning

Excellence-level practice questions

PRACTICE QUESTIONS

About this paper

- This is a **practice past paper**, written in the style of NCEA excellence questions. It is **not** an official NZQA paper, and it may or may not be similar to the actual exam.
- Questions aim at the excellence end of the standard, so expect them to be harder than an average exam question. Don't be discouraged if some take a while.
- **Questions 21 onwards are a bit harder** than the earlier questions in this paper.
- You may use the standard NZQA 91947 formula sheet, just as you would in the real exam.

How to use it

- Try each question on your own first, ideally under timed, exam-like conditions.
- Show full working and give clear final answers, as you would in the real exam.
- Only check the answers once you've made a proper attempt.

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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

1. A grain silo is made of a cylinder with a cone on top. Both have a radius of 3 m. The cylinder is 8 m tall and the cone is 4 m tall.

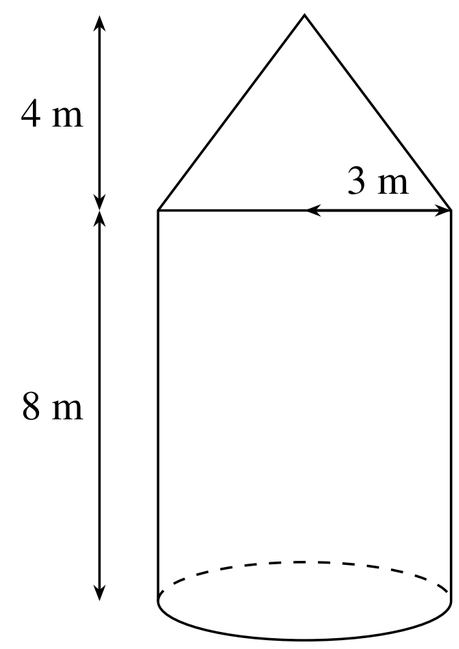


Diagram is NOT to scale

The silo is completely full of grain. 1 m^3 of grain weighs 760 kg.

One truck can carry 15 tonnes of grain in a single load.

Calculate the number of truck loads needed to empty the silo.

Justify your answer with clear mathematical reasoning.

2. Triangle ABC is equilateral. Its sides have lengths

$$\text{AB} = (2x + y)\text{ cm}, \qquad \text{BC} = (x + 3y)\text{ cm}, \qquad \text{CA} = (3x - y)\text{ cm}$$

The perimeter of the triangle is 60 cm.

(i)Find the values of x and y.

You must use an algebraic method, and show all steps of your working.

(ii)Find the exact area of triangle ABC.

3. The relationship between the x and y values in a sequence is shown in the table below.

x	1	2	3	4	5
y	5	14	29	50	77

(i)Find an equation that represents y for any given x-value.

Clearly show all steps of your working.

(ii)Find the value of x for which $y = 245$.

4. The diagram below shows a cuboid ABCDEFGH, where AB = 12 cm, BC = 9 cm and AE = 8 cm.

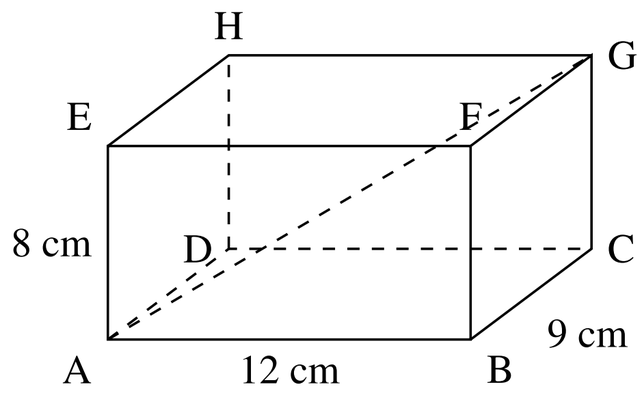


Diagram is NOT to scale

(i) Find the length of the diagonal AG.

Clearly show all steps of your working.

(ii) Find the size of the angle that AG makes with the base ABCD.

5. A container is in the shape of a cone, held with its point downwards. The cone has a height of 30 cm and a radius of 12 cm at the top.

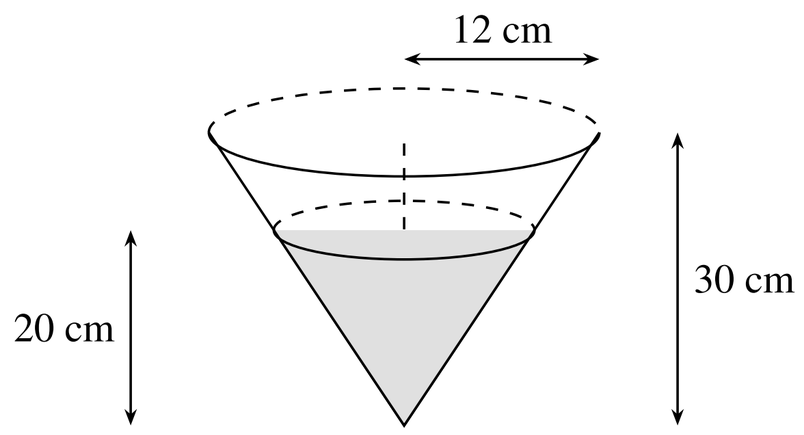


Diagram is NOT to scale

Water is poured into the container until it is 20 cm deep.

Find the exact volume of the water, giving your answer in terms of π.

Clearly show all steps of your working.

6. A rectangle has a width of $(x - 3)$ cm and a length of $(2x + 1)$ cm.

The perimeter of the rectangle is at least 40 cm, and its width is less than 10 cm.

Find all the integer values that x could take.

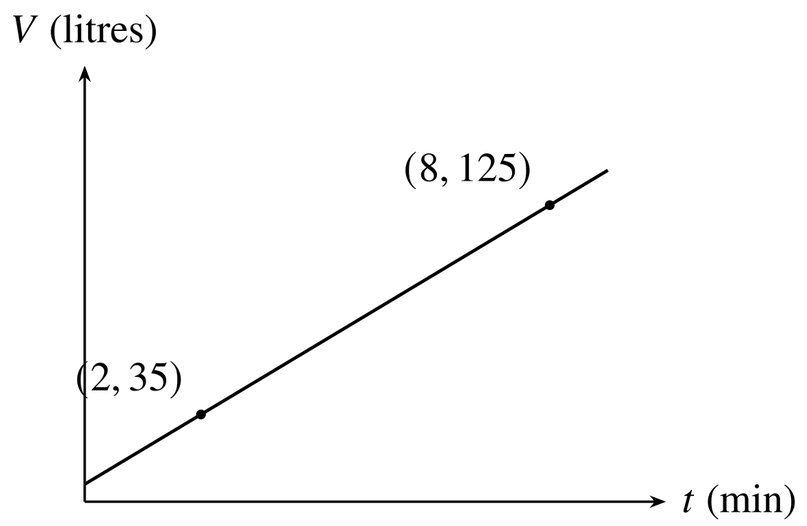
Justify your answer with clear mathematical reasoning.

7. (i)Prove that the sum of any three consecutive integers is always a multiple of 3.

(ii)Prove that the sum of any four consecutive integers is **never** a multiple of 4.

Justify your answer with clear mathematical reasoning.

8. Water is pumped into a tank at a constant rate. The graph below shows the volume of water, V litres, in the tank after t minutes. The graph is a straight line passing through the points $(2, 35)$ and $(8, 125)$.



(i) Find the rate, in litres per minute, at which the water is being pumped in, and explain how this rate relates to the graph.

(ii) Find the equation of the line, and state how much water was already in the tank when pumping began.

(iii) Find how long it takes for the tank to contain 500 litres.

9. A rectangular garden is 3 m longer than it is wide, and has an area of 40 m^2.

A path 1 m wide is laid all the way around the outside of the garden.

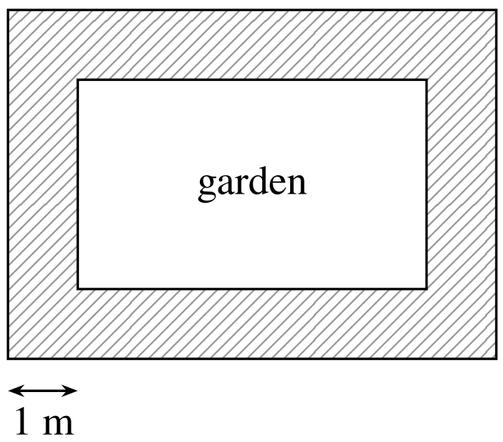


Diagram is NOT to scale

(i)Find the dimensions of the garden.

You must use an algebraic method, and show all steps of your working.

(ii)Find the area of the path.

10. ABCD is a rectangle with AB = 15 cm and AD = 8 cm.

The point P lies on AB, and angle ADP = 32°.

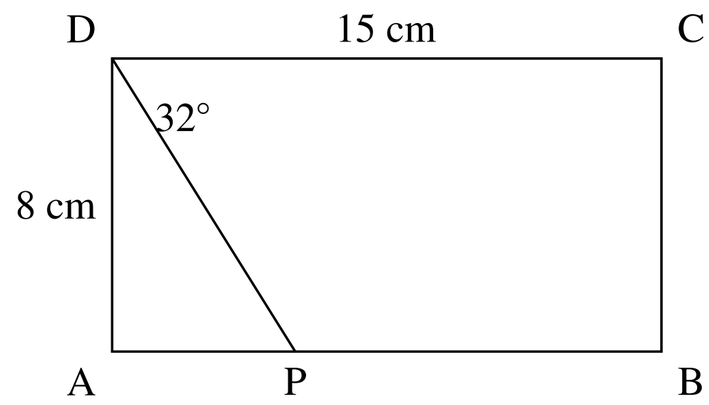


Diagram is NOT to scale

Find the length of PB.

Clearly show all steps of your working.

11. A solid toy is made from a cone joined to a hemisphere, as shown below.

The cone has a radius of 5 cm and a slant height of 13 cm. The hemisphere also has a radius of 5 cm.

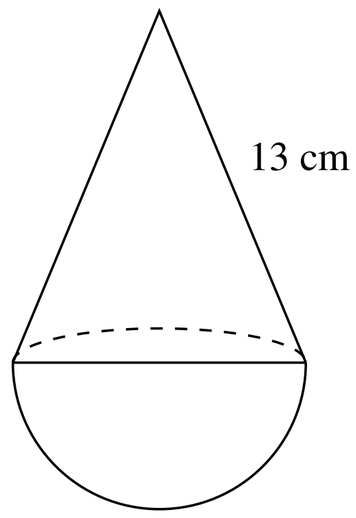


Diagram is NOT to scale

Find the exact total external surface area of the toy, giving your answer in terms of π.

Clearly show all steps of your working.

(Curved surface area of a cone = $\pi r L$, Surface area of a sphere = $4\pi r^2$)

12. At the Kōwhai Café, 3 coffees and 2 muffins cost \$25.40, and 5 coffees and 4 muffins cost \$44.00.

(i)Find the cost of one coffee and the cost of one muffin.

You must use an algebraic method, and show all steps of your working.

(ii)A different café charges \$$c$ for a coffee and \$$m$ for a muffin, where

$$3c + 2m = a \qquad \text{and} \qquad 5c + 4m = b$$

Find an expression for c in terms of a and b.

13. The relationship between the x and y values in a sequence is shown in the table below.

x	1	2	3	4
y	6	18	54	162

(i)Find an equation that represents y for any given x-value.

Clearly show all steps of your working.

(ii)Find the value of x for which $y = 4374$.

(iii)The graph of y could be drawn for all values of x.

Identify THREE different features of this graph, using your equation from part (i).

14. A vertical mast TM stands on horizontal ground, with its base at M.

The point A is due west of M, and the point B is due south of M.

From A, the angle of elevation of the top of the mast T is 38°. From B, the angle of elevation of T is 25°.
The distance AM is 40 m.

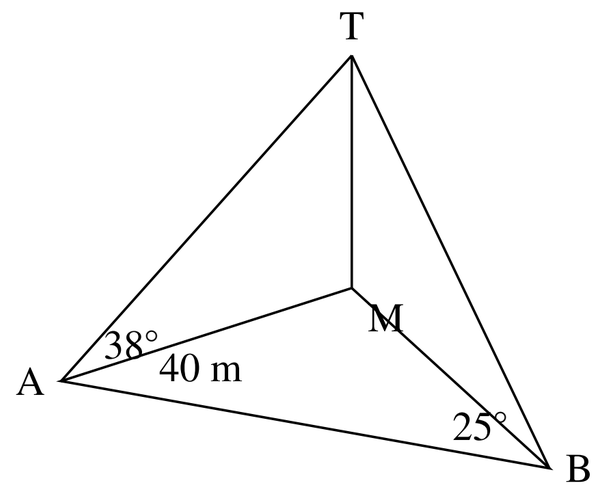


Diagram is NOT to scale

(i)Find the height of the mast TM.

(ii)Find the distance BM.

(iii)Find the distance AB.

Clearly show all steps of your working.

15. A farmer has 120 m of fencing to make a rectangular pen for her sheep.

One side of the pen is an existing straight wall, which needs no fencing. The other three sides are made from the 120 m of fencing.

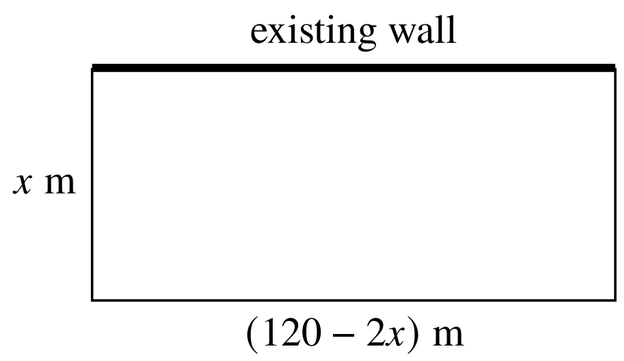


Diagram is NOT to scale

Find the dimensions of the pen that give the greatest possible area, and find this greatest area.

Justify your answer with clear mathematical reasoning.

16. Two cones are mathematically similar.

The smaller cone has a height of 12 cm and a volume of 200 cm^3.

The larger cone has a volume of 675 cm^3.

(i)Find the height of the larger cone.

Clearly show all steps of your working.

(ii)Find the ratio of the surface area of the smaller cone to the surface area of the larger cone.

17. (i)Simplify fully

$$\frac{\left(2x^3y^2\right)^3}{4x^5y}$$

(ii)Solve for x

$$2^{3x+1} \times 4^{x-2} = 8^{x+1}$$

You must use an algebraic method, and show all steps of your working.

18. The parabola M crosses the x-axis at $x = -2$ and $x = 6$, and crosses the y-axis at $(0, -12)$.

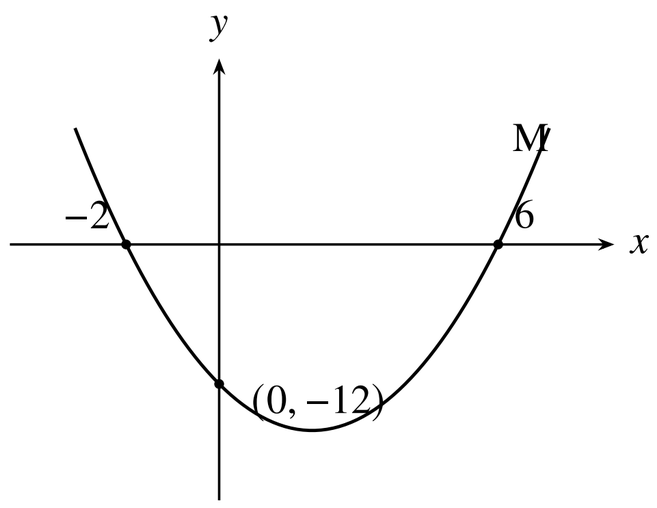


Diagram is NOT to scale

(i) Find the equation of the parabola M, in the form $y = ax^2 + bx + c$.

Justify your answer with clear mathematical reasoning.

(ii) Find the coordinates of the vertex of M.

(iii) Find the coordinates of the points where the line $y = 2x - 12$ meets the parabola M.

You must use an algebraic method, and show all steps of your working.

19. A container is made from a cylinder of radius r and height $3r$, with a hemisphere of radius r on top.

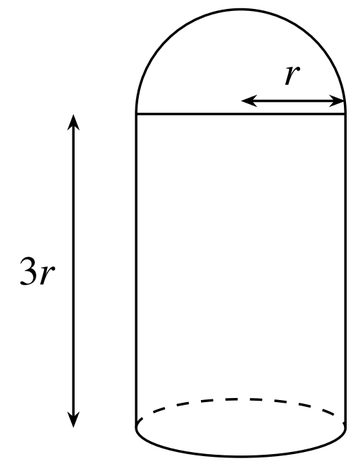


Diagram is NOT to scale

(i)Show that the total volume of the container is $\frac{11}{3}\pi r^3$.

Clearly show all steps of your working.

(ii)The container holds 2000 cm^3. Find the value of r.

20. A right-angled triangle has a hypotenuse of length $(x+5)$ cm, and the other two sides have lengths $(x-3)$ cm and $(x+4)$ cm.

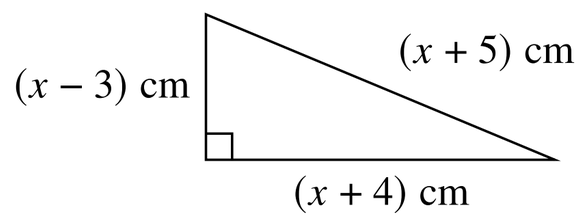


Diagram is NOT to scale

(i)Find the value of x.

You must use an algebraic method, and show all steps of your working.

(ii)Hence find the area of the triangle.

21. A regular polygon with n sides has all of its vertices on a circle of radius p, with centre Z.

The diagram below shows the case $n = 8$.

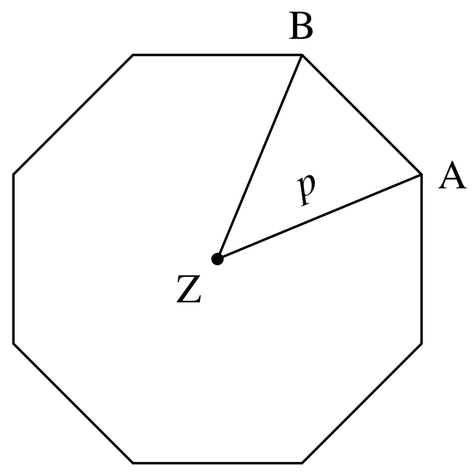


Diagram is NOT to scale

(i)By dividing the polygon into right-angled triangles, show that the area of the polygon is

$$n\,p^2 \sin\left(\frac{180^\circ}{n}\right)\cos\left(\frac{180^\circ}{n}\right)$$

Justify your answer with clear mathematical reasoning.

(ii)A regular octagon table top has $p = 120$ cm. Find its area.

22. A parabola has the equation

$$y = x^2 + kx + 9$$

where k is a constant.

(i) The parabola touches the x-axis at exactly one point. Find the possible values of k.
Justify your answer with clear mathematical reasoning.

(ii) Find all the values of k for which the parabola never crosses or touches the x-axis.

23. A pyramid has a square base of side $2a$, and each of its four slant edges has length $3a$.

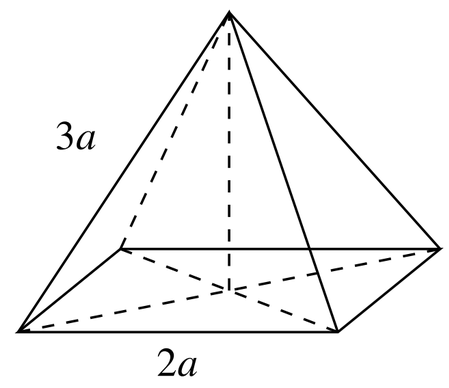


Diagram is NOT to scale

(i) Find the perpendicular height of the pyramid, in terms of a. Give your answer in its simplest exact form.

Clearly show all steps of your working.

(ii) Find the size of the angle between a slant edge and the base.

24. A pyramid has a square base of side 10 cm and a perpendicular height of 12 cm.

A cut is made parallel to the base, exactly halfway up the pyramid. The small pyramid on top is removed.

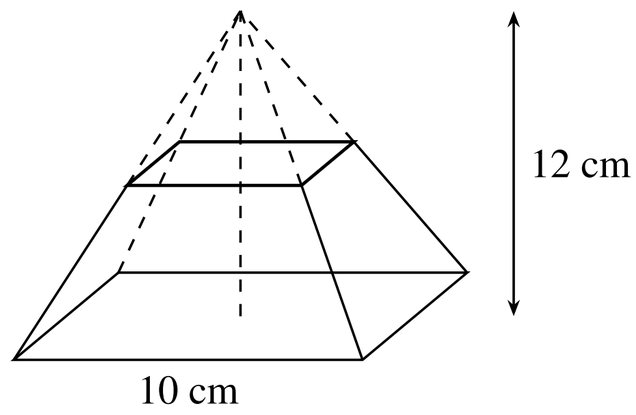


Diagram is NOT to scale

Find the volume of the remaining solid (the frustum).

Justify your answer with clear mathematical reasoning.

25. Prove that the difference between the squares of any two consecutive odd numbers is always a multiple of 8.

Justify your answer with clear mathematical reasoning.

26. Two straight lines have the equations

$$2x - 3y = 12 \qquad \text{and} \qquad 4x + y = 10$$

The lines meet at the point P.

(i) Find the coordinates of P.

You must use an algebraic method, and show all steps of your working.

(ii) The two lines and the y-axis form a triangle.

Find the area of this triangle.

27. A cone has a perpendicular height h and a base radius r.

The cone is cut parallel to its base, at a height of $\frac{1}{3}h$ measured down from the apex.

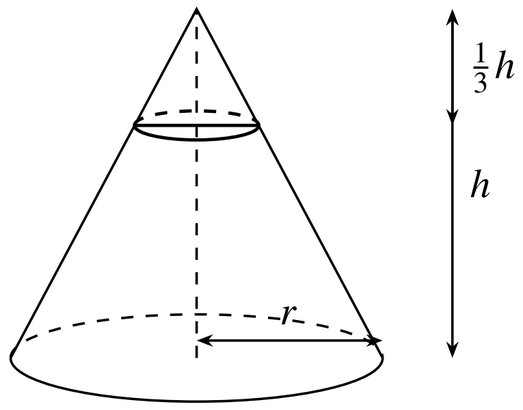


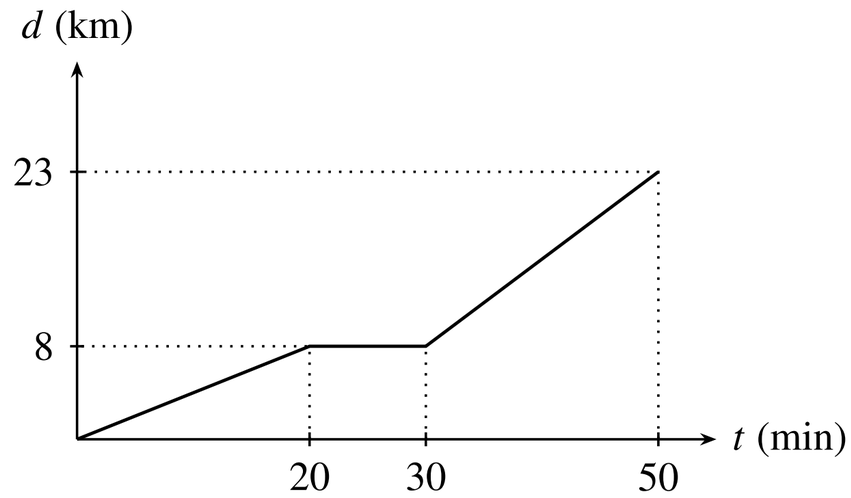
Diagram is NOT to scale

(i) Show that the volume of the small cone removed from the top is $\frac{1}{27}$ of the volume of the original cone.

Justify your answer with clear mathematical reasoning.

(ii) Find the ratio of the volume of the frustum (the part left behind) to the volume of the small cone.

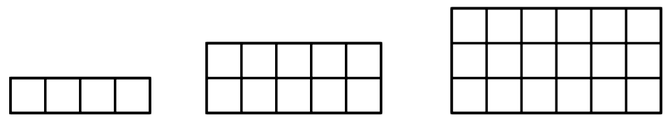
28. The graph below shows a cyclist's journey. The distance d is measured in kilometres, and the time t in minutes.



(i) Find the cyclist's speed, in $\text{km}\,\text{h}^{-1}$, during each of the three stages of the journey, and explain how each speed relates to the gradient of the graph.

(ii) Find the cyclist's average speed, in $\text{km}\,\text{h}^{-1}$, for the whole journey.

29. A sequence of patterns is made from unit squares. The first three patterns are shown below.



Pattern 1 Pattern 2 Pattern 3

Pattern 1 has 4 squares, Pattern 2 has 10 squares, and Pattern 3 has 18 squares.

(i)Find a formula for the number of squares, T, in Pattern n.

Justify your answer with clear mathematical reasoning.

(ii)Find which pattern is made from 460 squares.

You must use an algebraic method, and show all steps of your working.

30. A room is in the shape of a cuboid, 6 m long, 4 m wide and 3 m high.

A spider sits at the corner A on the floor, and a fly rests at the corner G, which is the corner diagonally opposite A.

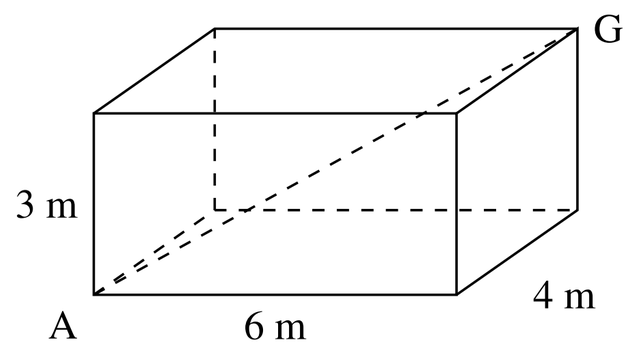


Diagram is NOT to scale

(i)Find the exact straight-line distance AG.

Clearly show all steps of your working.

(ii)Find the size of the angle that AG makes with the floor of the room.

31. Find all the values of x that satisfy both of the inequations

$$5(x-2) \geq 2x+11 \qquad \text{and} \qquad 3-2x > -17$$

State your answer as an inequality, and list all the integer values that x could take.

Justify your answer with clear mathematical reasoning.

32. A solid cylinder has radius r and height h.

The total surface area of the cylinder (including both circular ends) is equal to the surface area of a sphere of the same radius r.

(i) Show that $h = r$.

Justify your answer with clear mathematical reasoning.

(ii) Find the ratio of the volume of the cylinder to the volume of the sphere.

33. A straight ladder leans against a vertical wall. The foot of the ladder is 1.2 m from the base of the wall, and the ladder makes an angle of 70° with the horizontal ground.

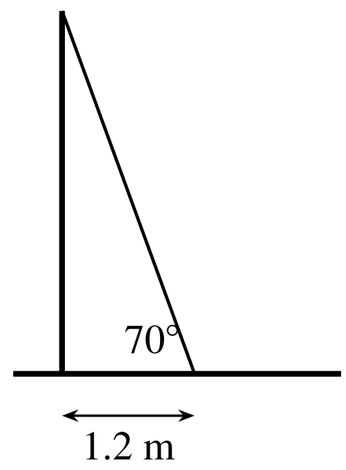


Diagram is NOT to scale

(i) Find the length of the ladder, and how far up the wall it reaches.

(ii) The foot of the ladder is then pulled away from the wall, until the ladder makes an angle of 55° with the ground.

How much further from the wall is the foot of the ladder, and how far down the wall has the top of the ladder slipped?

Clearly show all steps of your working.

34. The relationship between the x and y values in a sequence is shown in the table below.

x	0	1	2	3
y	7	11	19	35

(i)Find an equation that represents y for any given x-value.

Justify your answer with clear mathematical reasoning.

(ii)The graph of y could be drawn for all values of x.

Identify THREE different features of this graph, using your equation from part (i).

35. A tube is a cylinder that holds n tennis balls, stacked one on top of another. Each ball has radius r, and the balls fit tightly inside the tube, touching the sides, the base and the lid.

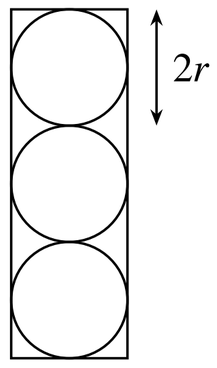


Diagram is NOT to scale

Show that the percentage of the tube's volume that is empty space is the same no matter what r and n are, and find this percentage.

Justify your answer with clear mathematical reasoning.

36. In triangle ABC, angle ABC = 90°.

The point D lies on AB and the point E lies on AC, so that DE is parallel to BC.

AD = 6 cm, DB = 4 cm, and DE = 9 cm.

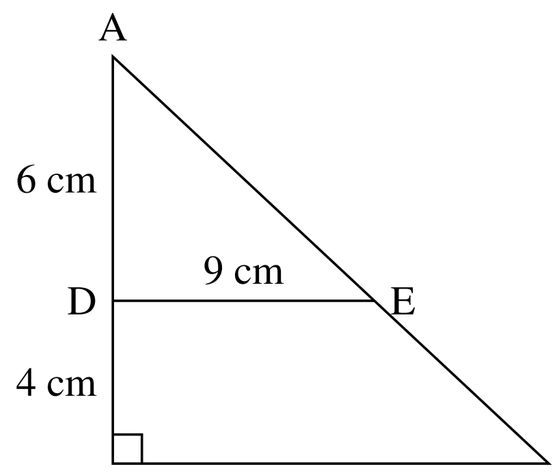


Diagram is NOT to scale

(i)Find the length of BC.

Justify your answer with clear mathematical reasoning.

(ii)Find the ratio of the area of triangle ADE to the area of the trapezium DBCE.

37. The line $y = 3x + k$ is a tangent to the parabola

$$y = x^2 + 5x + 10$$

That is, the line touches the parabola at exactly one point.

(i) Find the value of k.

You must use an algebraic method, and show all steps of your working.

(ii) Find the coordinates of the point where the line touches the parabola.

38. A school is hiring vehicles for a trip.

A van holds 12 students and costs $90 to hire. A bus holds 40 students and costs $250 to hire.

At least 200 students must travel, and the school can hire a maximum of 8 vehicles altogether.

Find the combination of vans and buses that costs the least, and state this least cost.

Justify your answer with clear mathematical reasoning.

39. A right-angled triangle has its two shorter sides of length a and b.

A square of side s is drawn inside the triangle, with one corner at the right angle, two sides lying along the two shorter sides of the triangle, and the opposite corner lying on the hypotenuse.

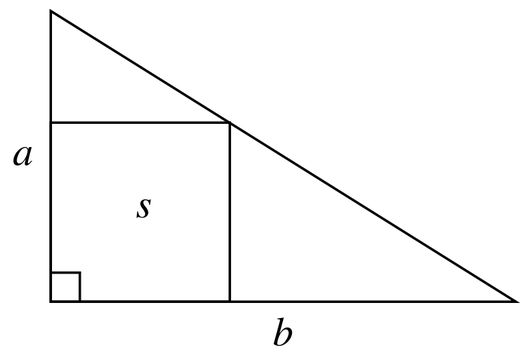


Diagram is NOT to scale

(i)Using similar triangles, show that

$$s = \frac{ab}{a+b}$$

Justify your answer with clear mathematical reasoning.

(ii)Find the value of s when $a = 6$ cm and $b = 12$ cm.

40. A cone has a base radius of 6 cm and a perpendicular height of 8 cm.

A sphere is placed inside the cone, so that the sphere touches the base of the cone and also touches the sloping side all the way around.

The diagram below shows the cross-section of the cone and sphere through the axis of the cone. In this cross-section the sphere becomes a circle that touches all three sides of the triangle.

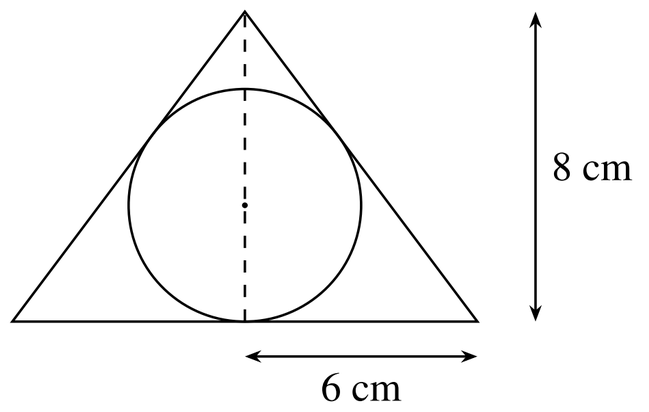


Diagram is NOT to scale

Find the radius of the sphere.

Justify your answer with clear mathematical reasoning.